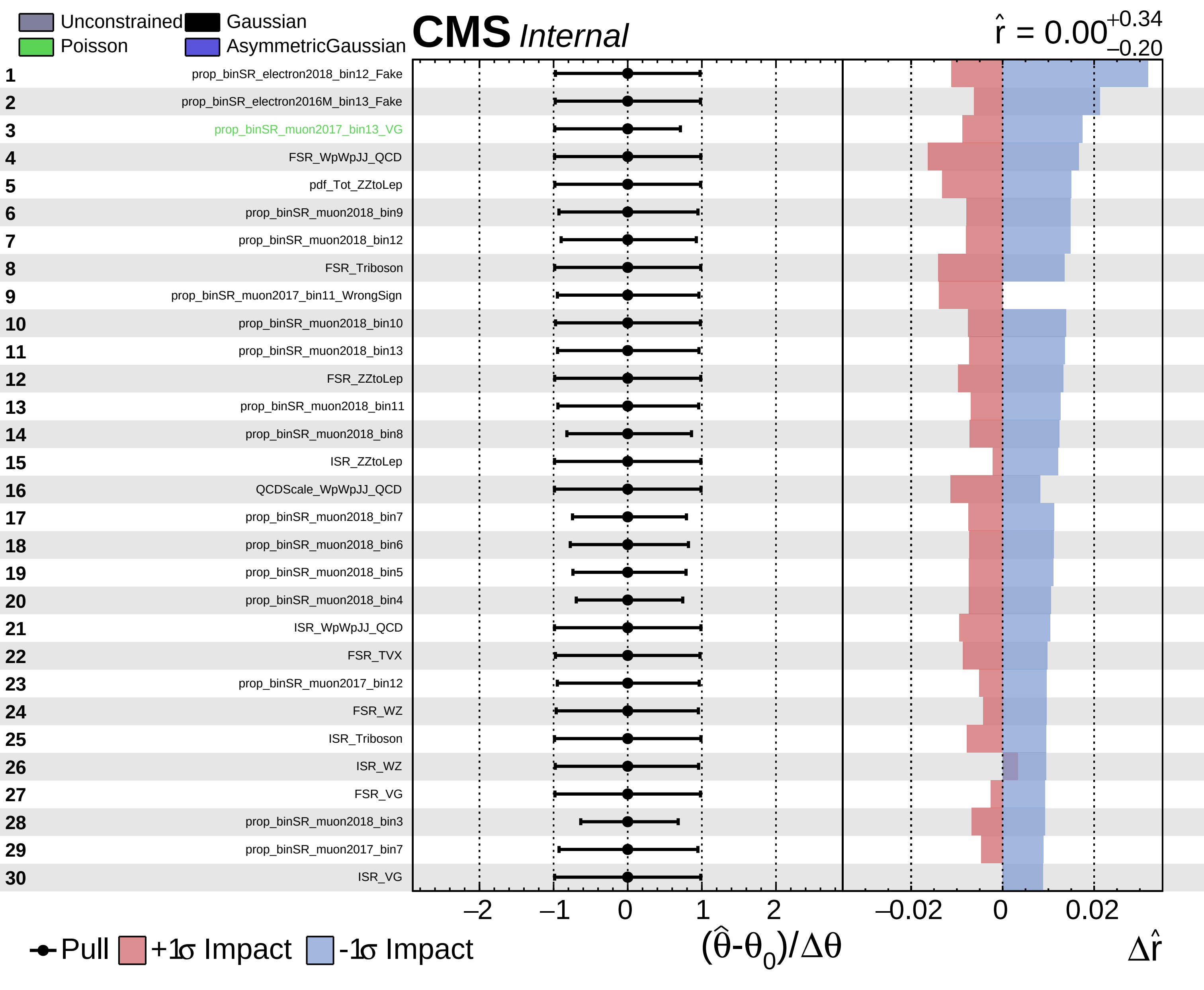
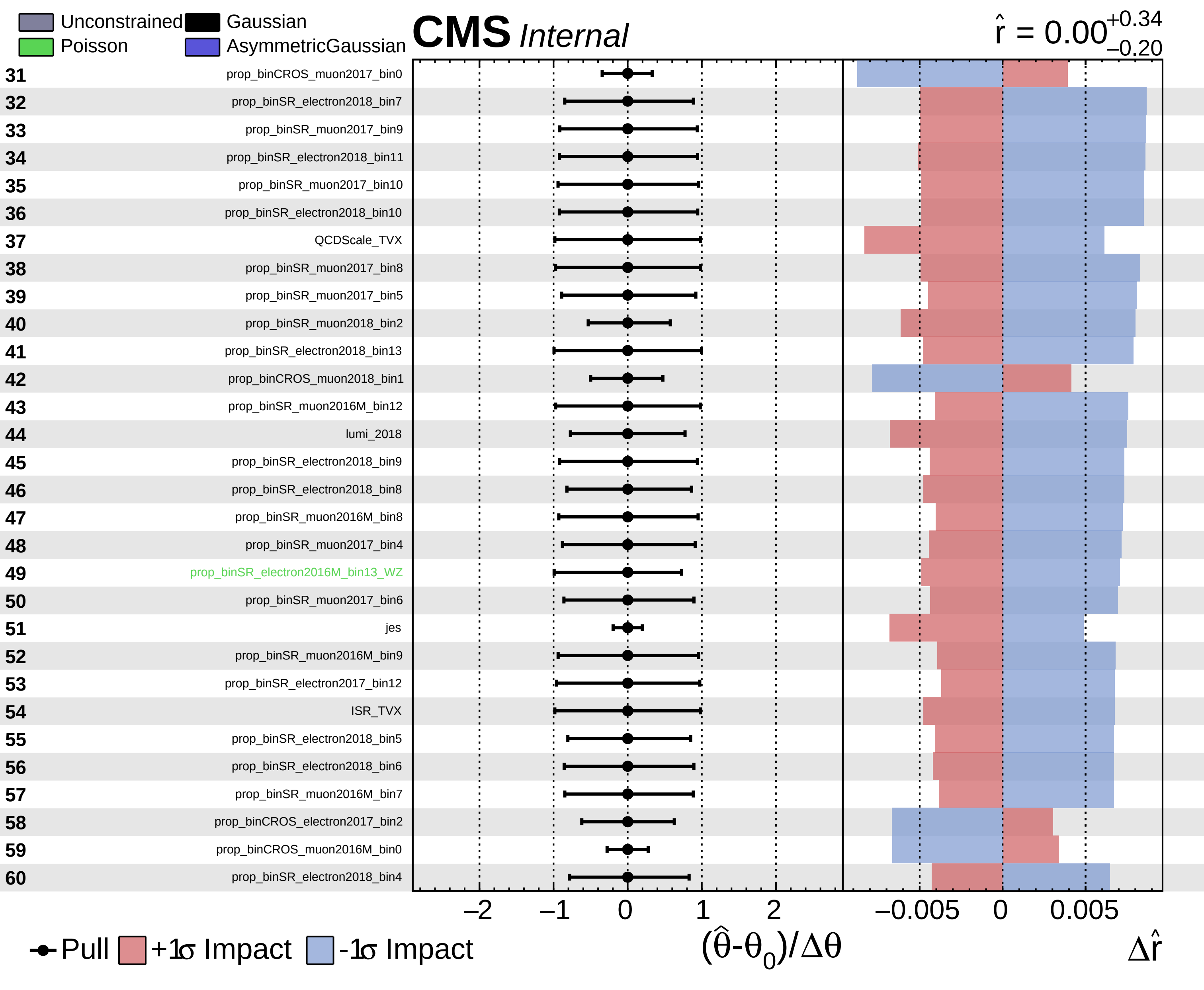
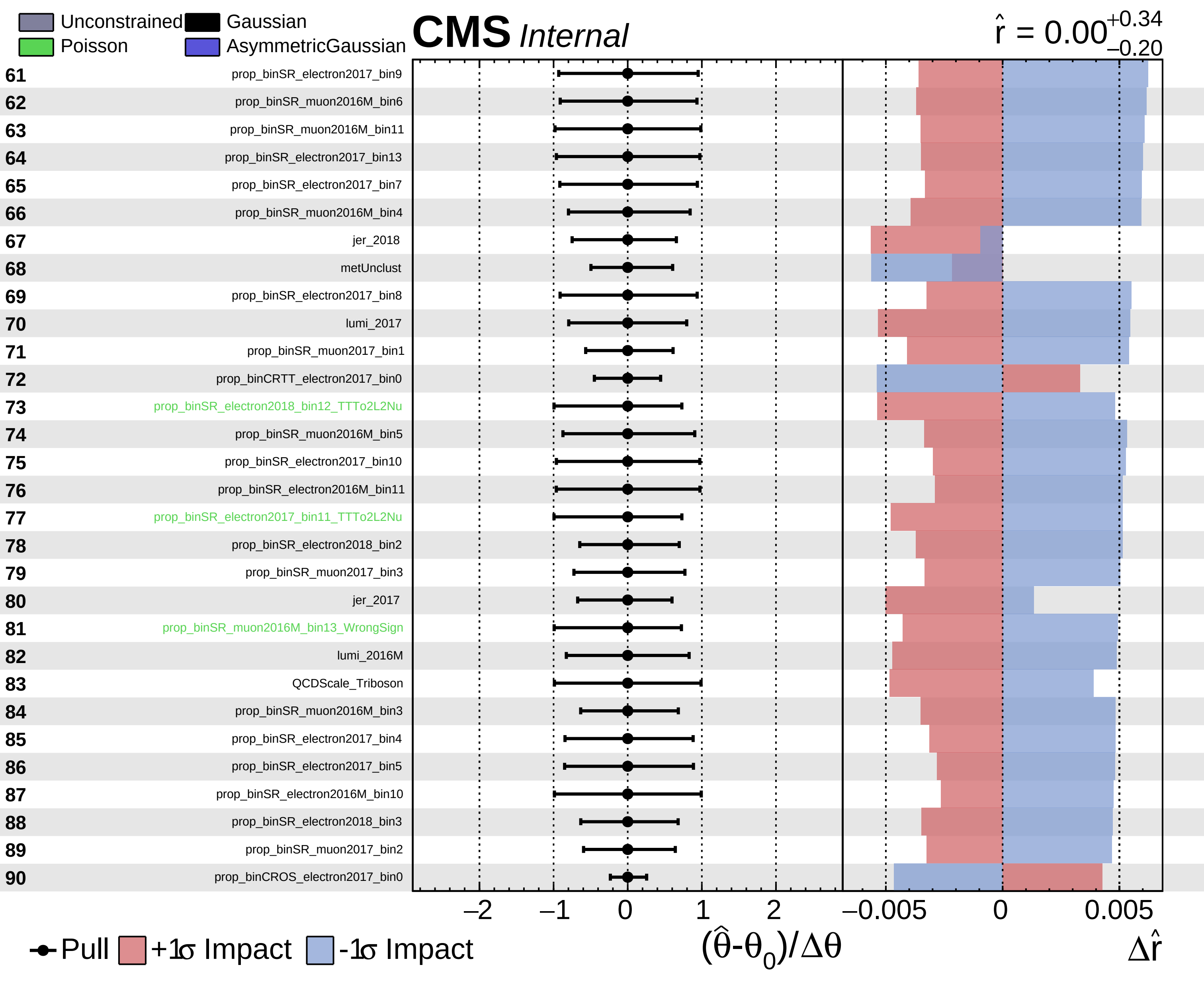


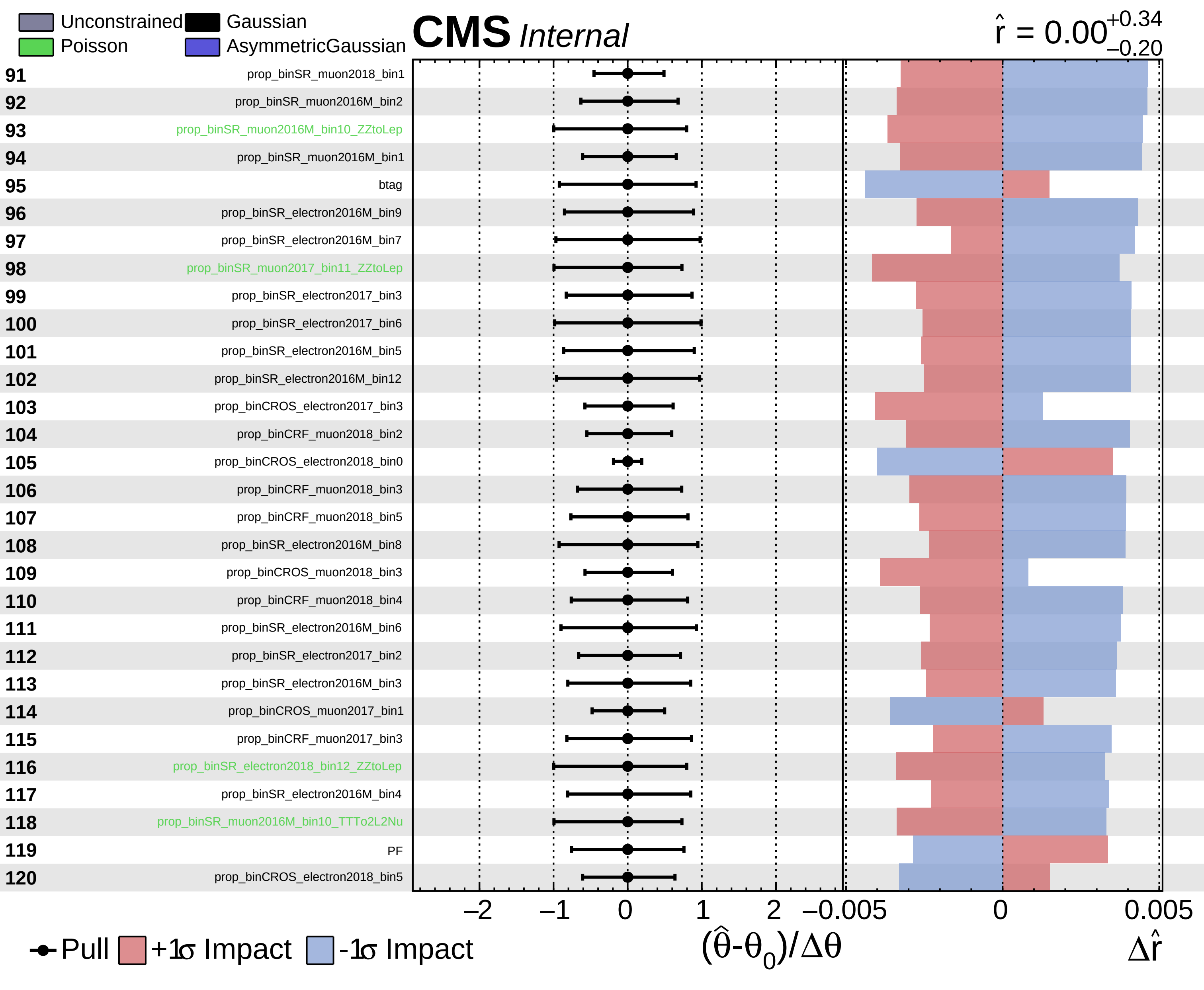
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
1 prop_binSR_electron2018_bin12_Fake
2 prop_binSR_electron2016M_bin13_Fake
3 prop_binSR_muon2017_bin13_VG
4 FSR_WpWpJJ_QCD
5 pdf_Tot_ZZtoLep
6 prop_binSR_muon2018_bin9
7 prop_binSR_muon2018_bin12
8 FSR_Triboson
9 prop_binSR_muon2017_bin11_WrongSign
10 prop_binSR_muon2018_bin10
11 prop_binSR_muon2018_bin13
12 FSR_ZZtoLep
13 prop_binSR_muon2018_bin11
14 prop_binSR_muon2018_bin8
15 ISR_ZZtoLep
16 QCDScale_WpWpJJ_QCD
17 prop_binSR_muon2018_bin7
18 prop_binSR_muon2018_bin6
19 prop_binSR_muon2018_bin5
20 prop_binSR_muon2018_bin4
21 ISR_WpWpJJ_QCD
22 FSR_TVX
23 prop_binSR_muon2017_bin12
24 FSR_WZ
25 ISR_Triboson
26 ISR_WZ
27 FSR_VG
28 prop_binSR_muon2018_bin3
29 prop_binSR_muon2017_bin7
30 ISR_VG
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

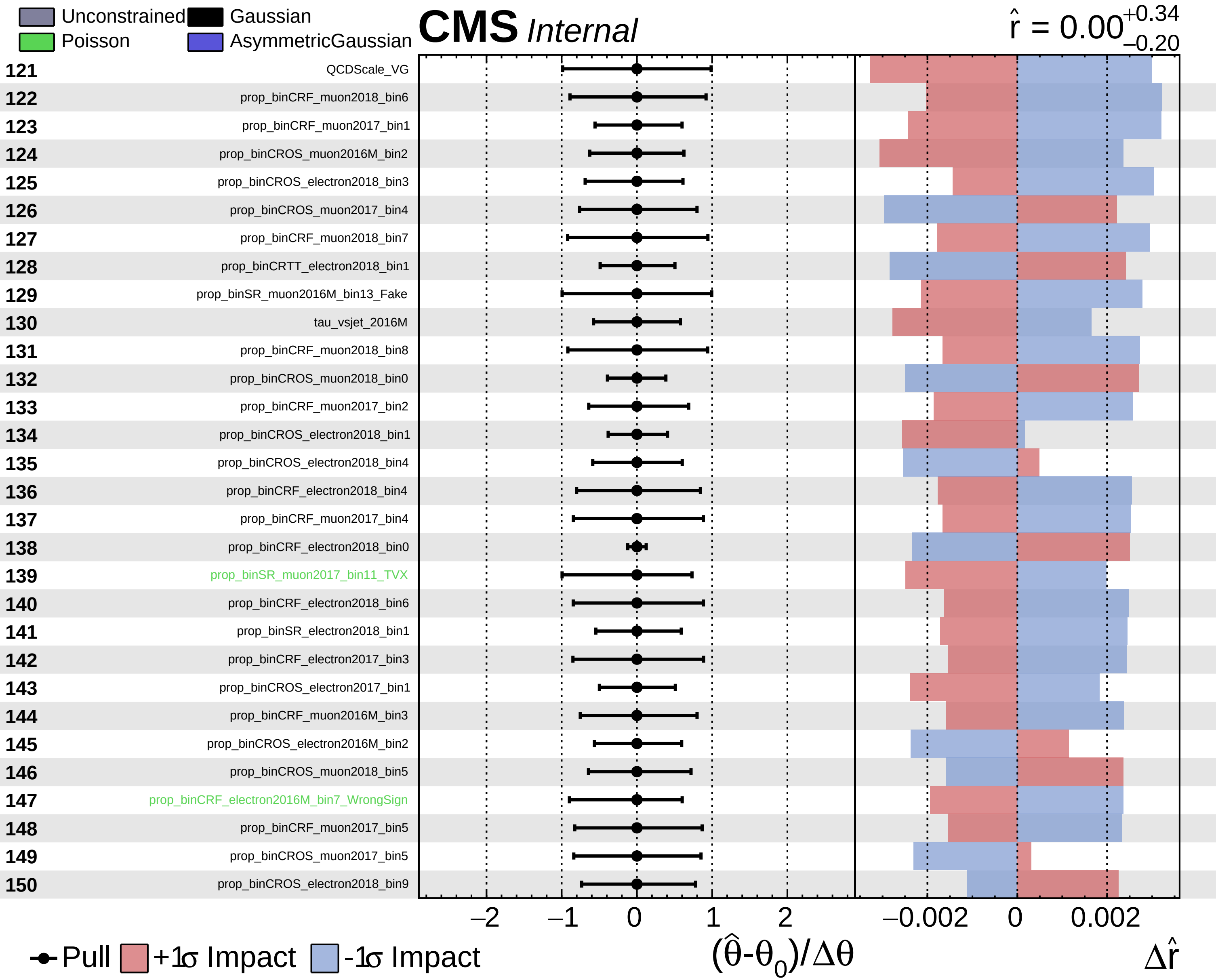


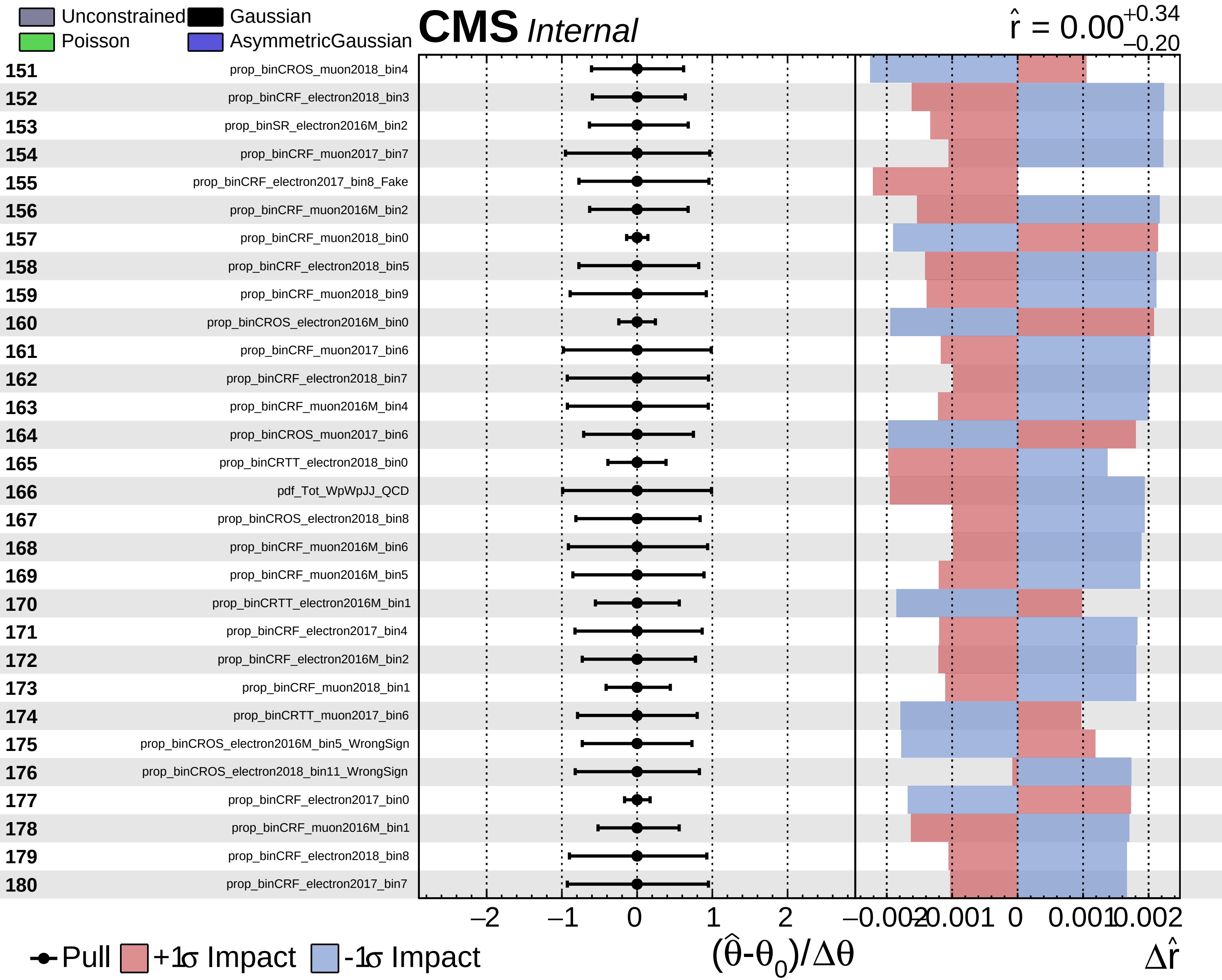
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
31 prop_binCROS_muon2017_bin0
32 prop_binSR_electron2018_bin7
33 prop_binSR_muon2017_bin9
34 prop_binSR_electron2018_bin11
35 prop_binSR_muon2017_bin10
36 prop_binSR_electron2018_bin10
37 QCDScale_TVX
38 prop_binSR_muon2017_bin8
39 prop_binSR_muon2017_bin5
40 prop_binSR_muon2018_bin2
41 prop_binSR_electron2018_bin13
42 prop_binCROS_muon2018_bin1
43 prop_binSR_muon2016M_bin12
44 lumi_2018
45 prop_binSR_electron2018_bin9
46 prop_binSR_electron2018_bin8
47 prop_binSR_muon2016M_bin8
48 prop_binSR_muon2017_bin4
49 prop_binSR_electron2016M_bin13_WZ
50 prop_binSR_muon2017_bin6
51 jes
52 prop_binSR_muon2016M_bin9
53 prop_binSR_electron2017_bin12
54 ISR_TVX
55 prop_binSR_electron2018_bin5
56 prop_binSR_electron2018_bin6
57 prop_binSR_muon2016M_bin7
58 prop_binCROS_electron2017_bin2
59 prop_binCROS_muon2016M_bin0
60 prop_binSR_electron2018_bin4
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

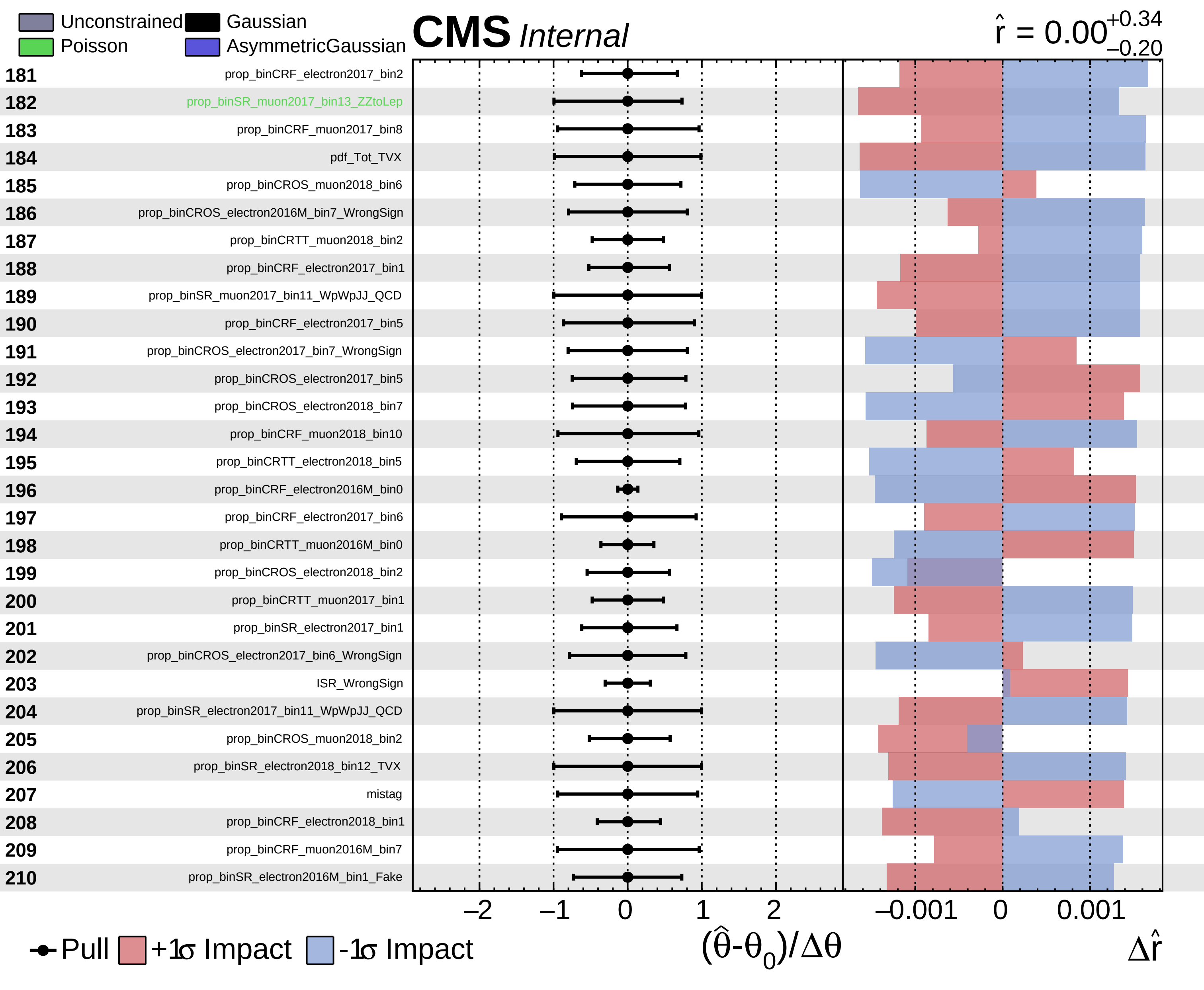


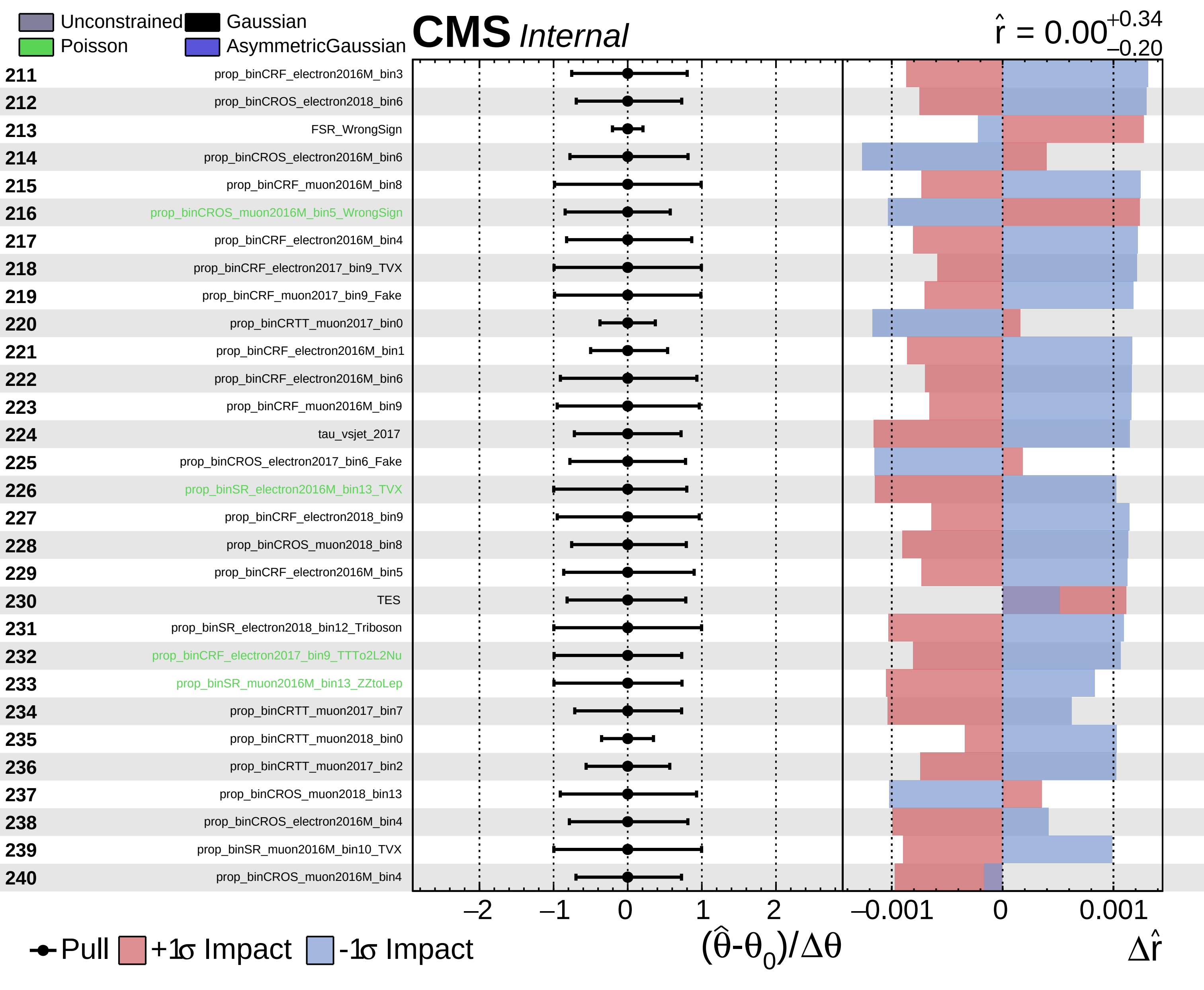
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
61 prop_binSR_electron2017_bin9
62 prop_binSR_muon2016M_bin6
63 prop_binSR_muon2016M_bin11
64 prop_binSR_electron2017_bin13
65 prop_binSR_electron2017_bin7
66 prop_binSR_muon2016M_bin4
67 jer_2018
68 metUnclust
69 prop_binSR_electron2017_bin8
70 lumi_2017
71 prop_binSR_muon2017_bin1
72 prop_binCRTT_electron2017_bin0
73 prop_binSR_electron2018_bin12_TTTo2L2Nu
74 prop_binSR_muon2016M_bin5
75 prop_binSR_electron2017_bin10
76 prop_binSR_electron2016M_bin11
77 prop_binSR_electron2017_bin11_TTTo2L2Nu
78 prop_binSR_electron2018_bin2
79 prop_binSR_muon2017_bin3
80 jer_2017
81 prop_binSR_muon2016M_bin13_WrongSign
82 lumi_2016M
83 QCDScale_Triboson
84 prop_binSR_muon2016M_bin3
85 prop_binSR_electron2017_bin4
86 prop_binSR_electron2017_bin5
87 prop_binSR_electron2016M_bin10
88 prop_binSR_electron2018_bin3
89 prop_binSR_muon2017_bin2
90 prop_binCROS_electron2017_bin0
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
91 prop_binSR_muon2018_bin1
92 prop_binSR_muon2016M_bin2
93 prop_binSR_muon2016M_bin10_ZZtoLep
94 prop_binSR_muon2016M_bin1
95 btag
96 prop_binSR_electron2016M_bin9
97 prop_binSR_electron2016M_bin7
98 prop_binSR_muon2017_bin11_ZZtoLep
99 prop_binSR_electron2017_bin3
100 prop_binSR_electron2017_bin6
101 prop_binSR_electron2016M_bin5
102 prop_binSR_electron2016M_bin12
103 prop_binCROS_electron2017_bin3
104 prop_binCRF_muon2018_bin2
105 prop_binCROS_electron2018_bin0
106 prop_binCRF_muon2018_bin3
107 prop_binCRF_muon2018_bin5
108 prop_binSR_electron2016M_bin8
109 prop_binCROS_muon2018_bin3
110 prop_binCRF_muon2018_bin4
111 prop_binSR_electron2016M_bin6
112 prop_binSR_electron2017_bin2
113 prop_binSR_electron2016M_bin3
114 prop_binCROS_muon2017_bin1
115 prop_binCRF_muon2017_bin3
116 prop_binSR_electron2018_bin12_ZZtoLep
117 prop_binSR_electron2016M_bin4
118 prop_binSR_muon2016M_bin10_TTTo2L2Nu
119 PF
120 prop_binCROS_electron2018_bin5
−2 −1 0 1 2 −0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

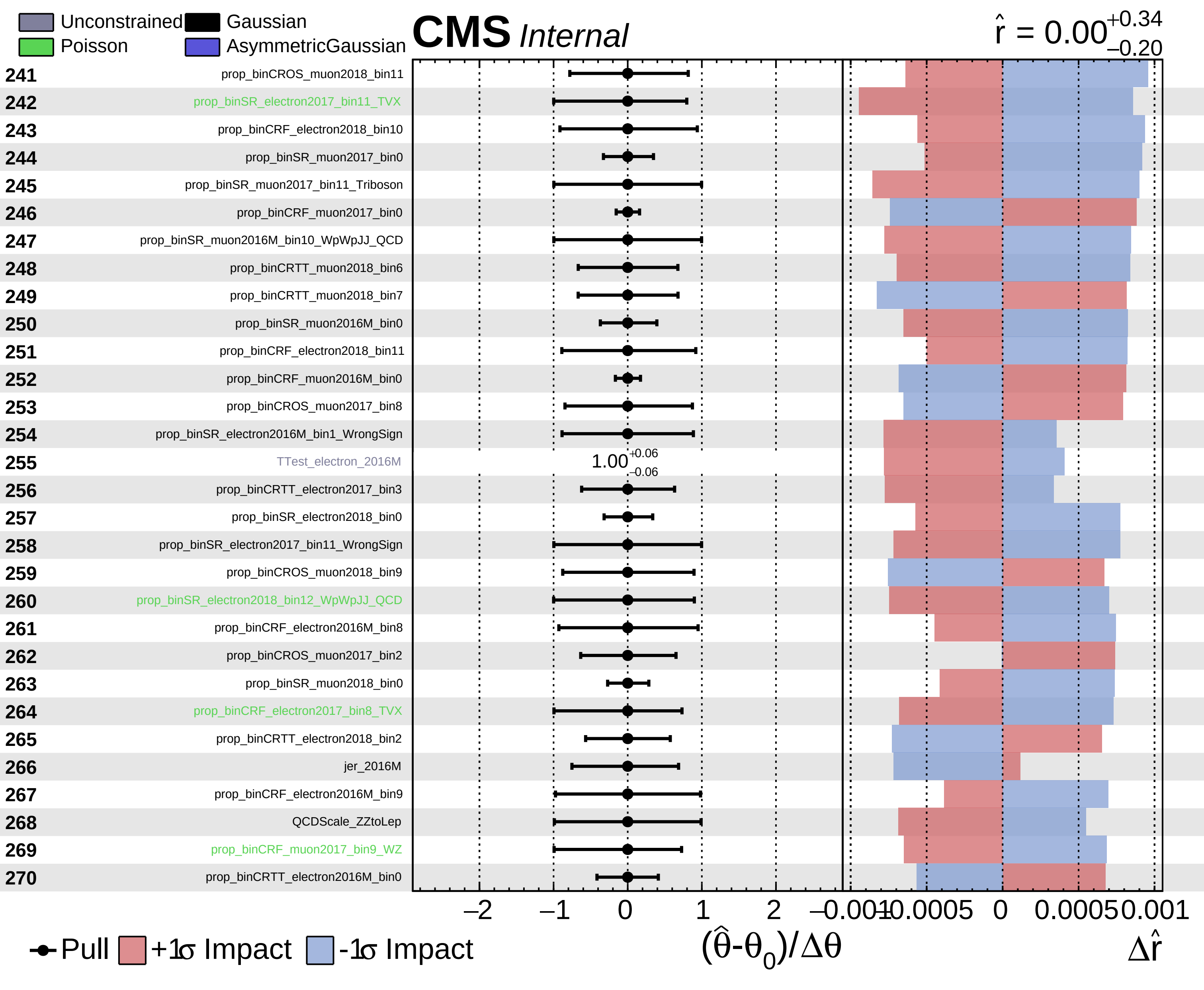

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
121 QCDScale_VG
122 prop_binCRF_muon2018_bin6
123 prop_binCRF_muon2017_bin1
124 prop_binCROS_muon2016M_bin2
125 prop_binCROS_electron2018_bin3
126 prop_binCROS_muon2017_bin4
127 prop_binCRF_muon2018_bin7
128 prop_binCRTT_electron2018_bin1
129 prop_binSR_muon2016M_bin13_Fake
130 tau_vsjet_2016M
131 prop_binCRF_muon2018_bin8
132 prop_binCROS_muon2018_bin0
133 prop_binCRF_muon2017_bin2
134 prop_binCROS_electron2018_bin1
135 prop_binCROS_electron2018_bin4
136 prop_binCRF_electron2018_bin4
137 prop_binCRF_muon2017_bin4
138 prop_binCRF_electron2018_bin0
139 prop_binSR_muon2017_bin11_TVX
140 prop_binCRF_electron2018_bin6
141 prop_binSR_electron2018_bin1
142 prop_binCRF_electron2017_bin3
143 prop_binCROS_electron2017_bin1
144 prop_binCRF_muon2016M_bin3
145 prop_binCROS_electron2016M_bin2
146 prop_binCROS_muon2018_bin5
147 prop_binCRF_electron2016M_bin7_WrongSign
148 prop_binCRF_muon2017_bin5
149 prop_binCROS_muon2017_bin5
150 prop_binCROS_electron2018_bin9
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
151 prop_binCROS_muon2018_bin4
152 prop_binCRF_electron2018_bin3
153 prop_binSR_electron2016M_bin2
154 prop_binCRF_muon2017_bin7
155 prop_binCRF_electron2017_bin8_Fake
156 prop_binCRF_muon2016M_bin2
157 prop_binCRF_muon2018_bin0
158 prop_binCRF_electron2018_bin5
159 prop_binCRF_muon2018_bin9
160 prop_binCROS_electron2016M_bin0
161 prop_binCRF_muon2017_bin6
162 prop_binCRF_electron2018_bin7
163 prop_binCRF_muon2016M_bin4
164 prop_binCROS_muon2017_bin6
165 prop_binCRTT_electron2018_bin0
166 pdf_Tot_WpWpJJ_QCD
167 prop_binCROS_electron2018_bin8
168 prop_binCRF_muon2016M_bin6
169 prop_binCRF_muon2016M_bin5
170 prop_binCRTT_electron2016M_bin1
171 prop_binCRF_electron2017_bin4
172 prop_binCRF_electron2016M_bin2
173 prop_binCRF_muon2018_bin1
174 prop_binCRTT_muon2017_bin6
175 prop_binCROS_electron2016M_bin5_WrongSign
176 prop_binCROS_electron2018_bin11_WrongSign
177 prop_binCRF_electron2017_bin0
178 prop_binCRF_muon2016M_bin1
179 prop_binCRF_electron2018_bin8
180 prop_binCRF_electron2017_bin7
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

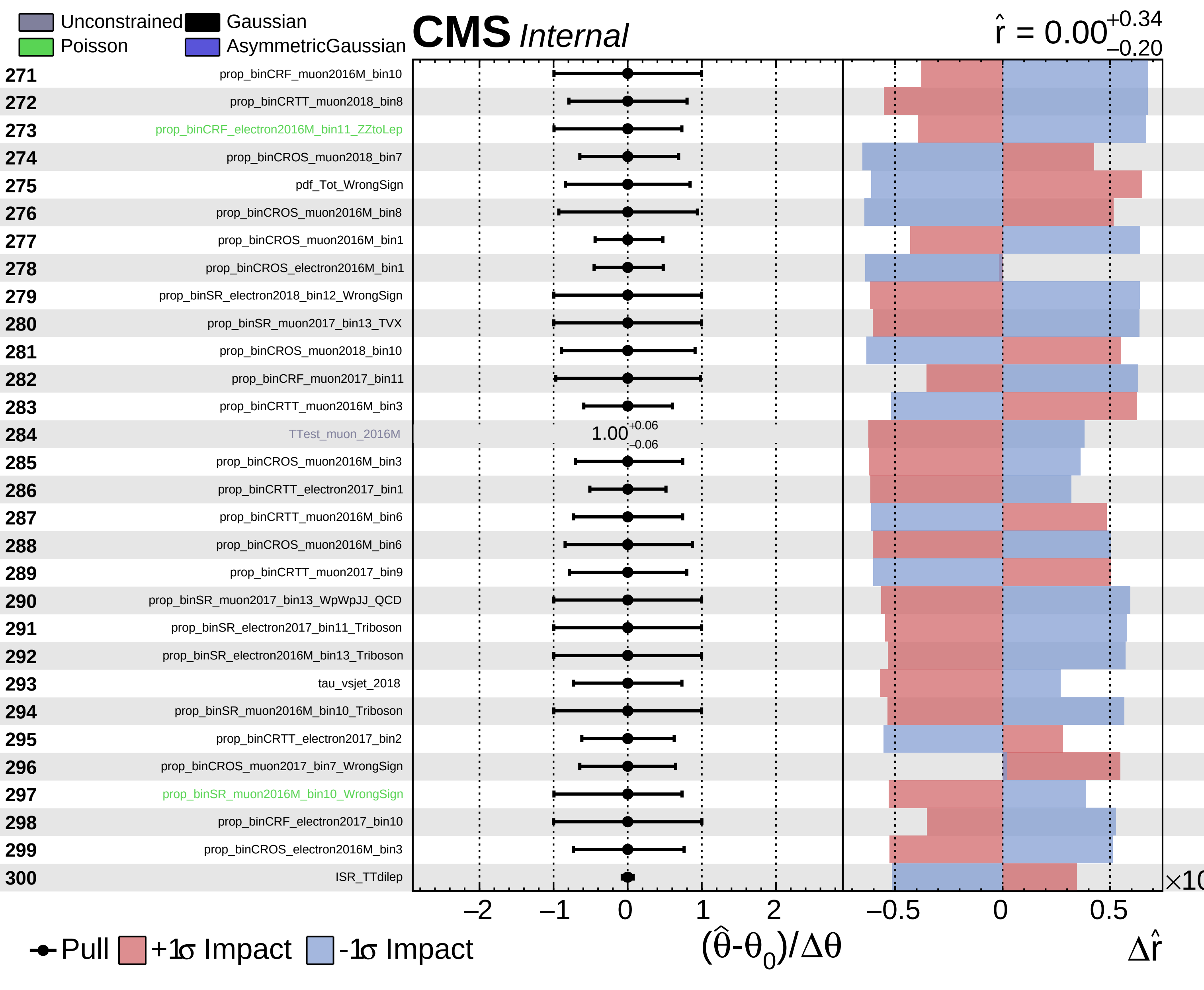

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
181 prop_binCRF_electron2017_bin2
182 prop_binSR_muon2017_bin13_ZZtoLep
183 prop_binCRF_muon2017_bin8
184 pdf_Tot_TVX
185 prop_binCROS_muon2018_bin6
186 prop_binCROS_electron2016M_bin7_WrongSign
187 prop_binCRTT_muon2018_bin2
188 prop_binCRF_electron2017_bin1
189 prop_binSR_muon2017_bin11_WpWpJJ_QCD
190 prop_binCRF_electron2017_bin5
191 prop_binCROS_electron2017_bin7_WrongSign
192 prop_binCROS_electron2017_bin5
193 prop_binCROS_electron2018_bin7
194 prop_binCRF_muon2018_bin10
195 prop_binCRTT_electron2018_bin5
196 prop_binCRF_electron2016M_bin0
197 prop_binCRF_electron2017_bin6
198 prop_binCRTT_muon2016M_bin0
199 prop_binCROS_electron2018_bin2
200 prop_binCRTT_muon2017_bin1
201 prop_binSR_electron2017_bin1
202 prop_binCROS_electron2017_bin6_WrongSign
203 ISR_WrongSign
204 prop_binSR_electron2017_bin11_WpWpJJ_QCD
205 prop_binCROS_muon2018_bin2
206 prop_binSR_electron2018_bin12_TVX
207 mistag
208 prop_binCRF_electron2018_bin1
209 prop_binCRF_muon2016M_bin7
210 prop_binSR_electron2016M_bin1_Fake
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



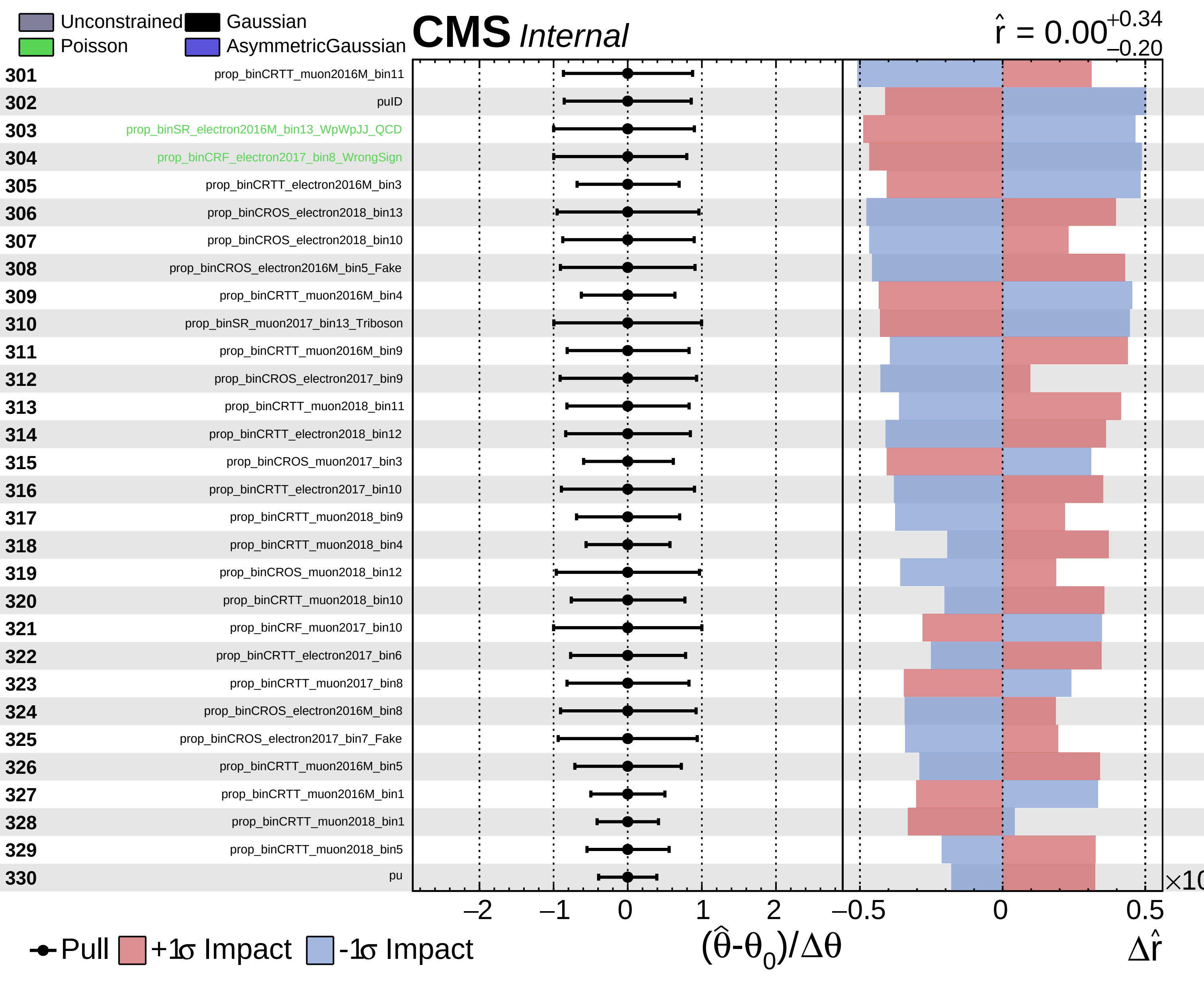
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
211 prop_binCRF_electron2016M_bin3
212 prop_binCROS_electron2018_bin6
213 FSR_WrongSign
214 prop_binCROS_electron2016M_bin6
215 prop_binCRF_muon2016M_bin8
216 prop_binCROS_muon2016M_bin5_WrongSign
217 prop_binCRF_electron2016M_bin4
218 prop_binCRF_electron2017_bin9_TVX
219 prop_binCRF_muon2017_bin9_Fake
220 prop_binCRTT_muon2017_bin0
221 prop_binCRF_electron2016M_bin1
222 prop_binCRF_electron2016M_bin6
223 prop_binCRF_muon2016M_bin9
224 tau_vsjet_2017
225 prop_binCROS_electron2017_bin6_Fake
226 prop_binSR_electron2016M_bin13_TVX
227 prop_binCRF_electron2018_bin9
228 prop_binCROS_muon2018_bin8
229 prop_binCRF_electron2016M_bin5
230 TES
231 prop_binSR_electron2018_bin12_Triboson
232 prop_binCRF_electron2017_bin9_TTTo2L2Nu
233 prop_binSR_muon2016M_bin13_ZZtoLep
234 prop_binCRTT_muon2017_bin7
235 prop_binCRTT_muon2018_bin0
236 prop_binCRTT_muon2017_bin2
237 prop_binCROS_muon2018_bin13
238 prop_binCROS_electron2016M_bin4
239 prop_binSR_muon2016M_bin10_TVX
240 prop_binCROS_muon2016M_bin4
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



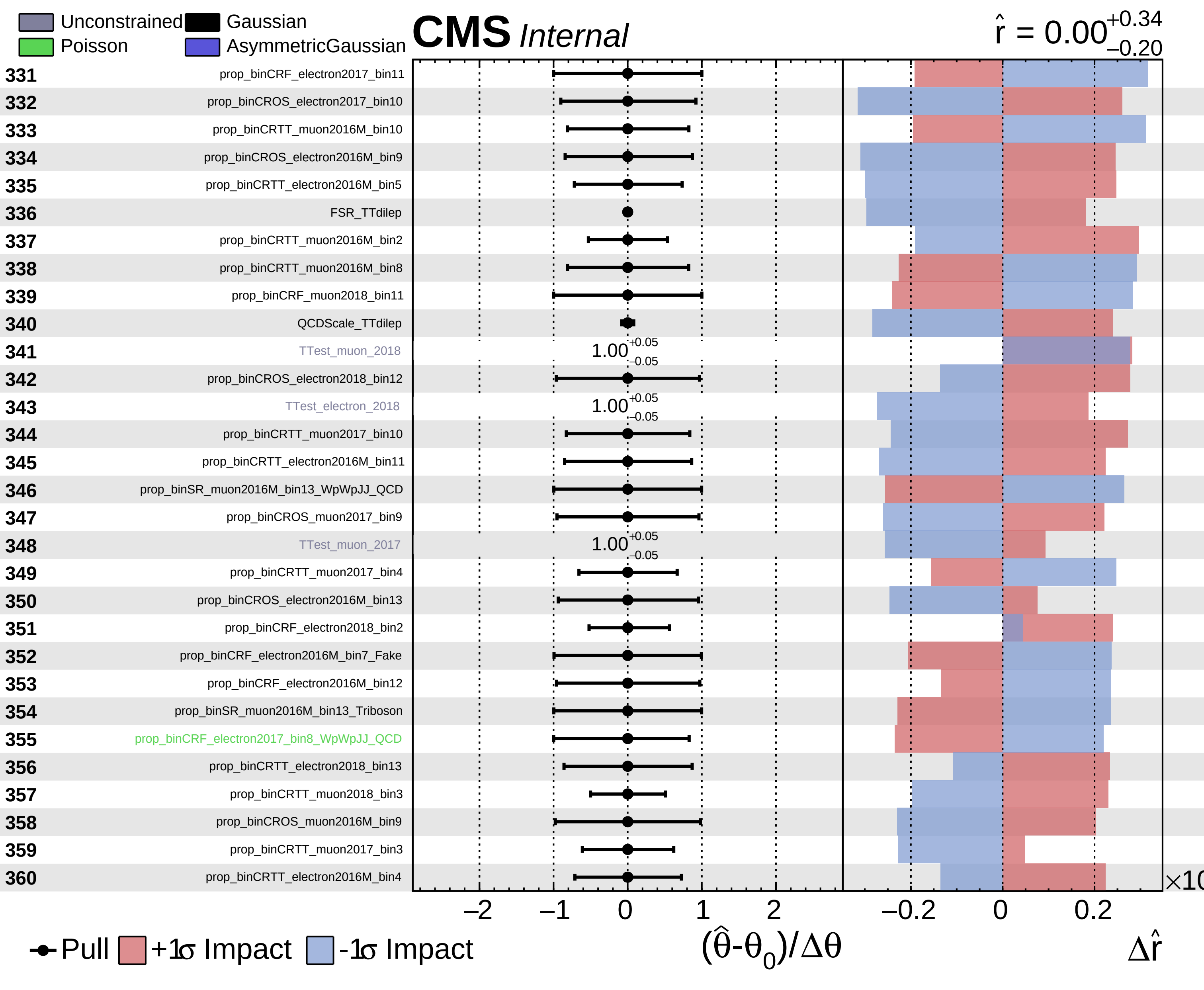
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
241 prop_binCROS_muon2018_bin11
242 prop_binSR_electron2017_bin11_TVX
243 prop_binCRF_electron2018_bin10
244 prop_binSR_muon2017_bin0
245 prop_binSR_muon2017_bin11_Triboson
246 prop_binCRF_muon2017_bin0
247 prop_binSR_muon2016M_bin10_WpWpJJ_QCD
248 prop_binCRTT_muon2018_bin6
249 prop_binCRTT_muon2018_bin7
250 prop_binSR_muon2016M_bin0
251 prop_binCRF_electron2018_bin11
252 prop_binCRF_muon2016M_bin0
253 prop_binCROS_muon2017_bin8
254 prop_binSR_electron2016M_bin1_WrongSign
255 TTest_electron_2016M
1.00 +0.06 −0.06
256 prop_binCRTT_electron2017_bin3
257 prop_binSR_electron2018_bin0
258 prop_binSR_electron2017_bin11_WrongSign
259 prop_binCROS_muon2018_bin9
260 prop_binSR_electron2018_bin12_WpWpJJ_QCD
261 prop_binCRF_electron2016M_bin8
262 prop_binCROS_muon2017_bin2
263 prop_binSR_muon2018_bin0
264 prop_binCRF_electron2017_bin8_TVX
265 prop_binCRTT_electron2018_bin2
266 jer_2016M
267 prop_binCRF_electron2016M_bin9
268 QCDScale_ZZtoLep
269 prop_binCRF_muon2017_bin9_WZ
270 prop_binCRTT_electron2016M_bin0
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

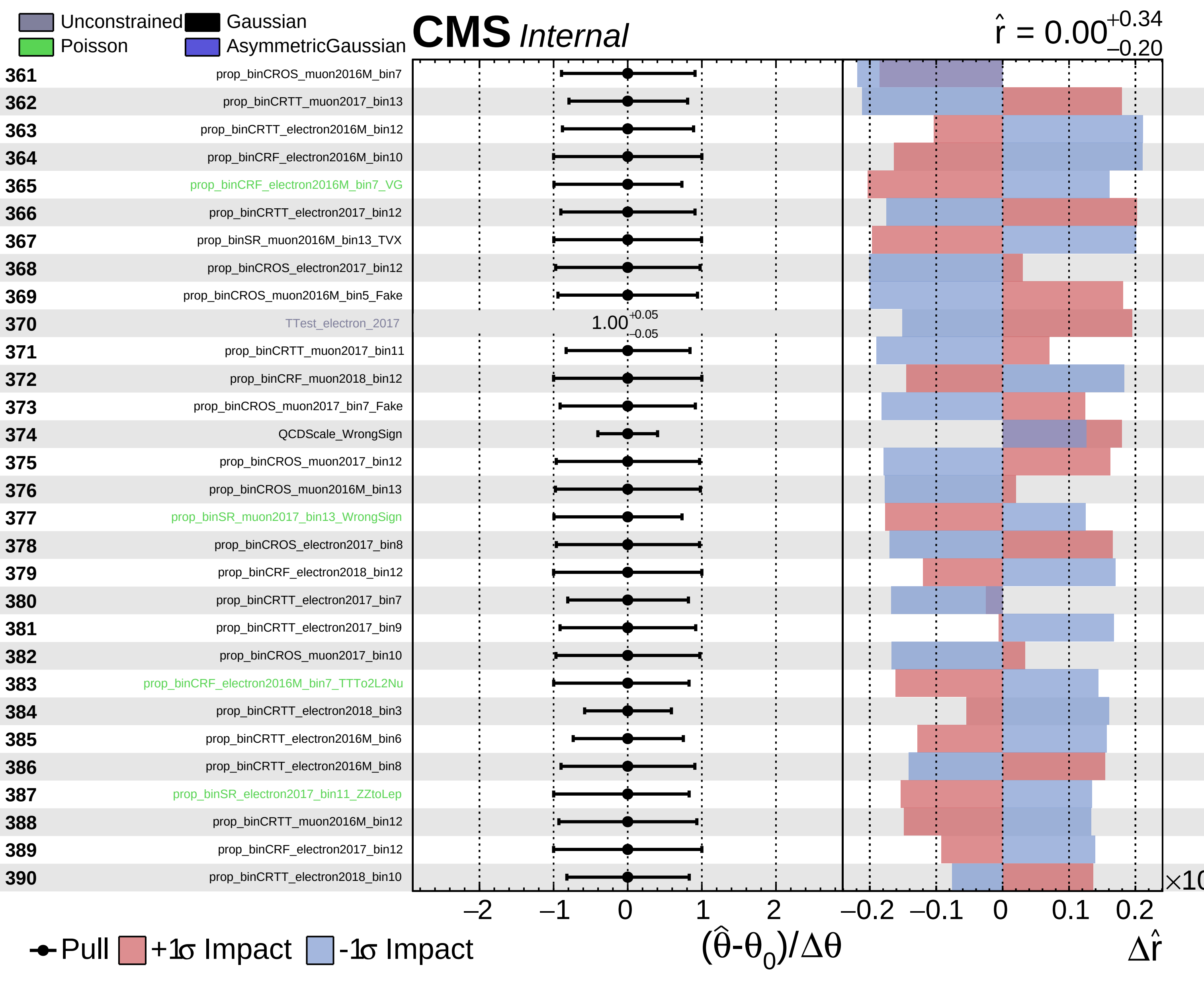


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
271 prop_binCRF_muon2016M_bin10
272 prop_binCRTT_muon2018_bin8
273 prop_binCRF_electron2016M_bin11_ZZtoLep
274 prop_binCROS_muon2018_bin7
275 pdf_Tot_WrongSign
276 prop_binCROS_muon2016M_bin8
277 prop_binCROS_muon2016M_bin1
278 prop_binCROS_electron2016M_bin1
279 prop_binSR_electron2018_bin12_WrongSign
280 prop_binSR_muon2017_bin13_TVX
281 prop_binCROS_muon2018_bin10
282 prop_binCRF_muon2017_bin11
283 prop_binCRTT_muon2016M_bin3
284 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
285 prop_binCROS_muon2016M_bin3
286 prop_binCRTT_electron2017_bin1
287 prop_binCRTT_muon2016M_bin6
288 prop_binCROS_muon2016M_bin6
289 prop_binCRTT_muon2017_bin9
290 prop_binSR_muon2017_bin13_WpWpJJ_QCD
291 prop_binSR_electron2017_bin11_Triboson
292 prop_binSR_electron2016M_bin13_Triboson
293 tau_vsjet_2018
294 prop_binSR_muon2016M_bin10_Triboson
295 prop_binCRTT_electron2017_bin2
296 prop_binCROS_muon2017_bin7_WrongSign
297 prop_binSR_muon2016M_bin10_WrongSign
298 prop_binCRF_electron2017_bin10
299 prop_binCROS_electron2016M_bin3
300 ISR_TTdilep
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

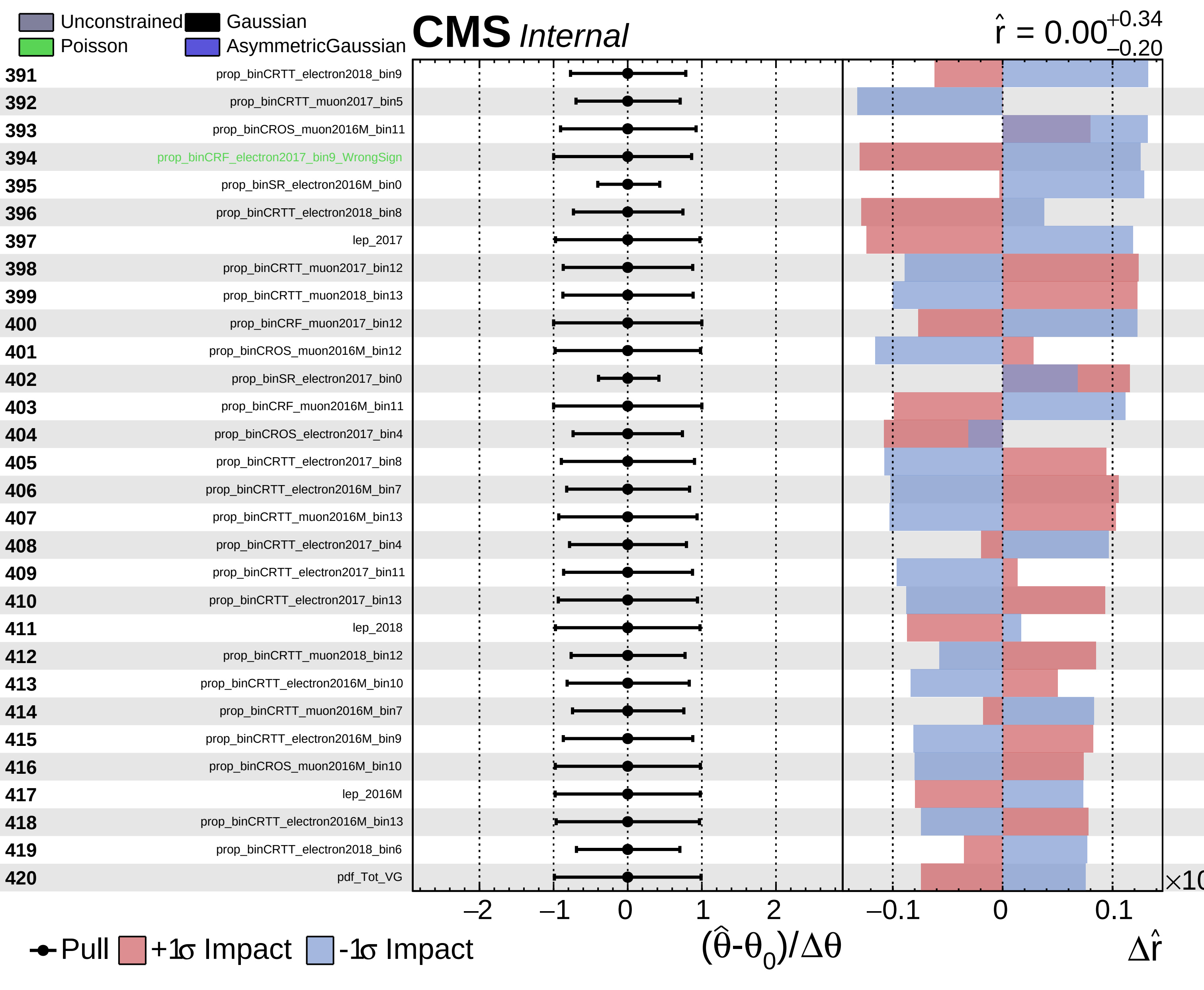


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
301 prop_binCRTT_muon2016M_bin11
302 puID
303 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
304 prop_binCRF_electron2017_bin8_WrongSign
305 prop_binCRTT_electron2016M_bin3
306 prop_binCROS_electron2018_bin13
307 prop_binCROS_electron2018_bin10
308 prop_binCROS_electron2016M_bin5_Fake
309 prop_binCRTT_muon2016M_bin4
310 prop_binSR_muon2017_bin13_Triboson
311 prop_binCRTT_muon2016M_bin9
312 prop_binCROS_electron2017_bin9
313 prop_binCRTT_muon2018_bin11
314 prop_binCRTT_electron2018_bin12
315 prop_binCROS_muon2017_bin3
316 prop_binCRTT_electron2017_bin10
317 prop_binCRTT_muon2018_bin9
318 prop_binCRTT_muon2018_bin4
319 prop_binCROS_muon2018_bin12
320 prop_binCRTT_muon2018_bin10
321 prop_binCRF_muon2017_bin10
322 prop_binCRTT_electron2017_bin6
323 prop_binCRTT_muon2017_bin8
324 prop_binCROS_electron2016M_bin8
325 prop_binCROS_electron2017_bin7_Fake
326 prop_binCRTT_muon2016M_bin5
327 prop_binCRTT_muon2016M_bin1
328 prop_binCRTT_muon2018_bin1
329 prop_binCRTT_muon2018_bin5
330 pu
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

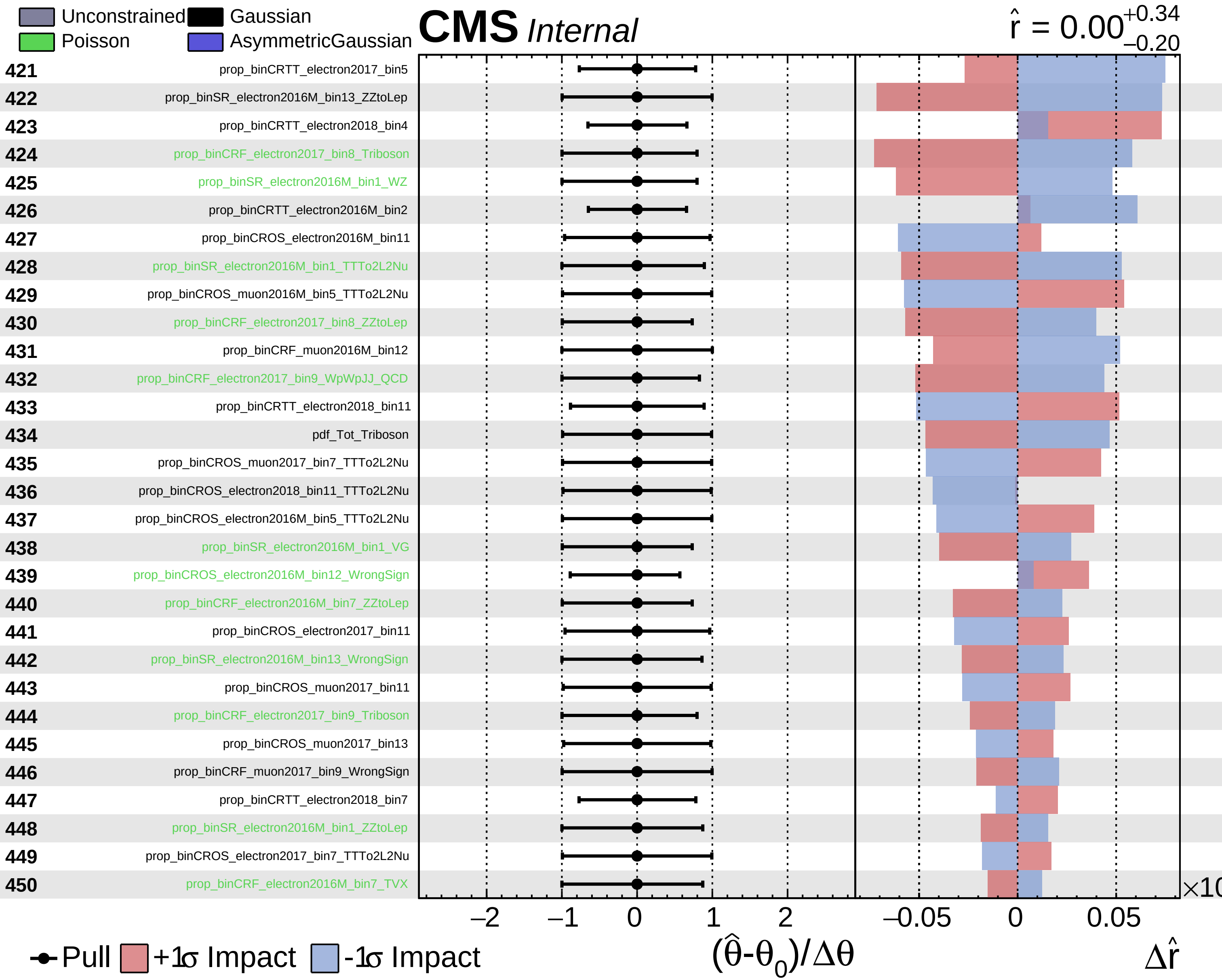

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
331 prop_binCRF_electron2017_bin11
332 prop_binCROS_electron2017_bin10
333 prop_binCRTT_muon2016M_bin10
334 prop_binCROS_electron2016M_bin9
335 prop_binCRTT_electron2016M_bin5
336 FSR_TTdilep
337 prop_binCRTT_muon2016M_bin2
338 prop_binCRTT_muon2016M_bin8
339 prop_binCRF_muon2018_bin11
340 QCDScale_TTdilep
341 TTest_muon_2018
1.00 +0.05 −0.05
342 prop_binCROS_electron2018_bin12
343 TTest_electron_2018
1.00 +0.05 −0.05
344 prop_binCRTT_muon2017_bin10
345 prop_binCRTT_electron2016M_bin11
346 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
347 prop_binCROS_muon2017_bin9
348 TTest_muon_2017
1.00 +0.05 −0.05
349 prop_binCRTT_muon2017_bin4
350 prop_binCROS_electron2016M_bin13
351 prop_binCRF_electron2018_bin2
352 prop_binCRF_electron2016M_bin7_Fake
353 prop_binCRF_electron2016M_bin12
354 prop_binSR_muon2016M_bin13_Triboson
355 prop_binCRF_electron2017_bin8_WpWpJJ_QCD
356 prop_binCRTT_electron2018_bin13
357 prop_binCRTT_muon2018_bin3
358 prop_binCROS_muon2016M_bin9
359 prop_binCRTT_muon2017_bin3
360 prop_binCRTT_electron2016M_bin4
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



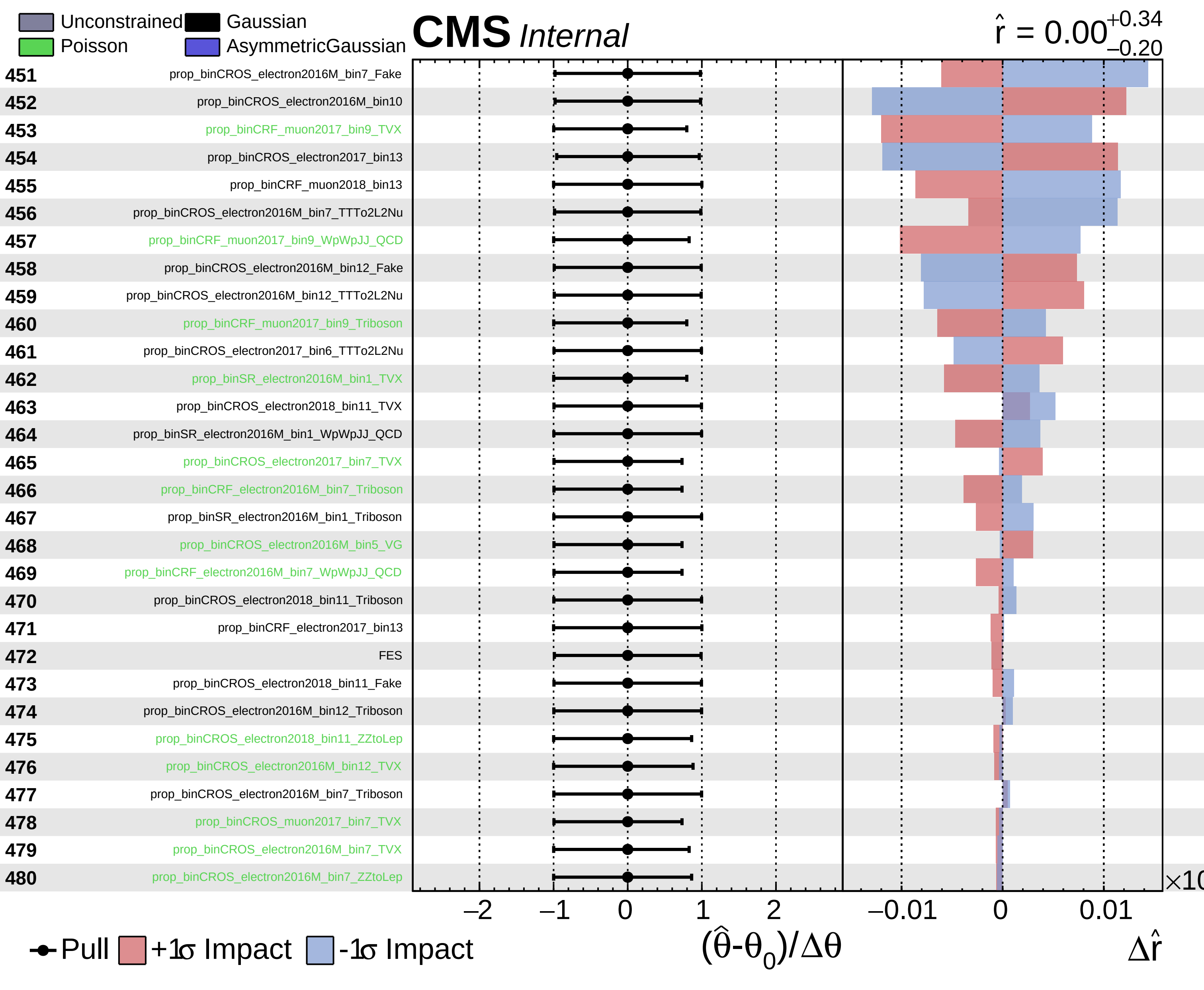
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
361 prop_binCROS_muon2016M_bin7
362 prop_binCRTT_muon2017_bin13
363 prop_binCRTT_electron2016M_bin12
364 prop_binCRF_electron2016M_bin10
365 prop_binCRF_electron2016M_bin7_VG
366 prop_binCRTT_electron2017_bin12
367 prop_binSR_muon2016M_bin13_TVX
368 prop_binCROS_electron2017_bin12
369 prop_binCROS_muon2016M_bin5_Fake
370 TTest_electron_2017
1.00 +0.05 −0.05
371 prop_binCRTT_muon2017_bin11
372 prop_binCRF_muon2018_bin12
373 prop_binCROS_muon2017_bin7_Fake
374 QCDScale_WrongSign
375 prop_binCROS_muon2017_bin12
376 prop_binCROS_muon2016M_bin13
377 prop_binSR_muon2017_bin13_WrongSign
378 prop_binCROS_electron2017_bin8
379 prop_binCRF_electron2018_bin12
380 prop_binCRTT_electron2017_bin7
381 prop_binCRTT_electron2017_bin9
382 prop_binCROS_muon2017_bin10
383 prop_binCRF_electron2016M_bin7_TTTo2L2Nu
384 prop_binCRTT_electron2018_bin3
385 prop_binCRTT_electron2016M_bin6
386 prop_binCRTT_electron2016M_bin8
387 prop_binSR_electron2017_bin11_ZZtoLep
388 prop_binCRTT_muon2016M_bin12
389 prop_binCRF_electron2017_bin12
390 prop_binCRTT_electron2018_bin10
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

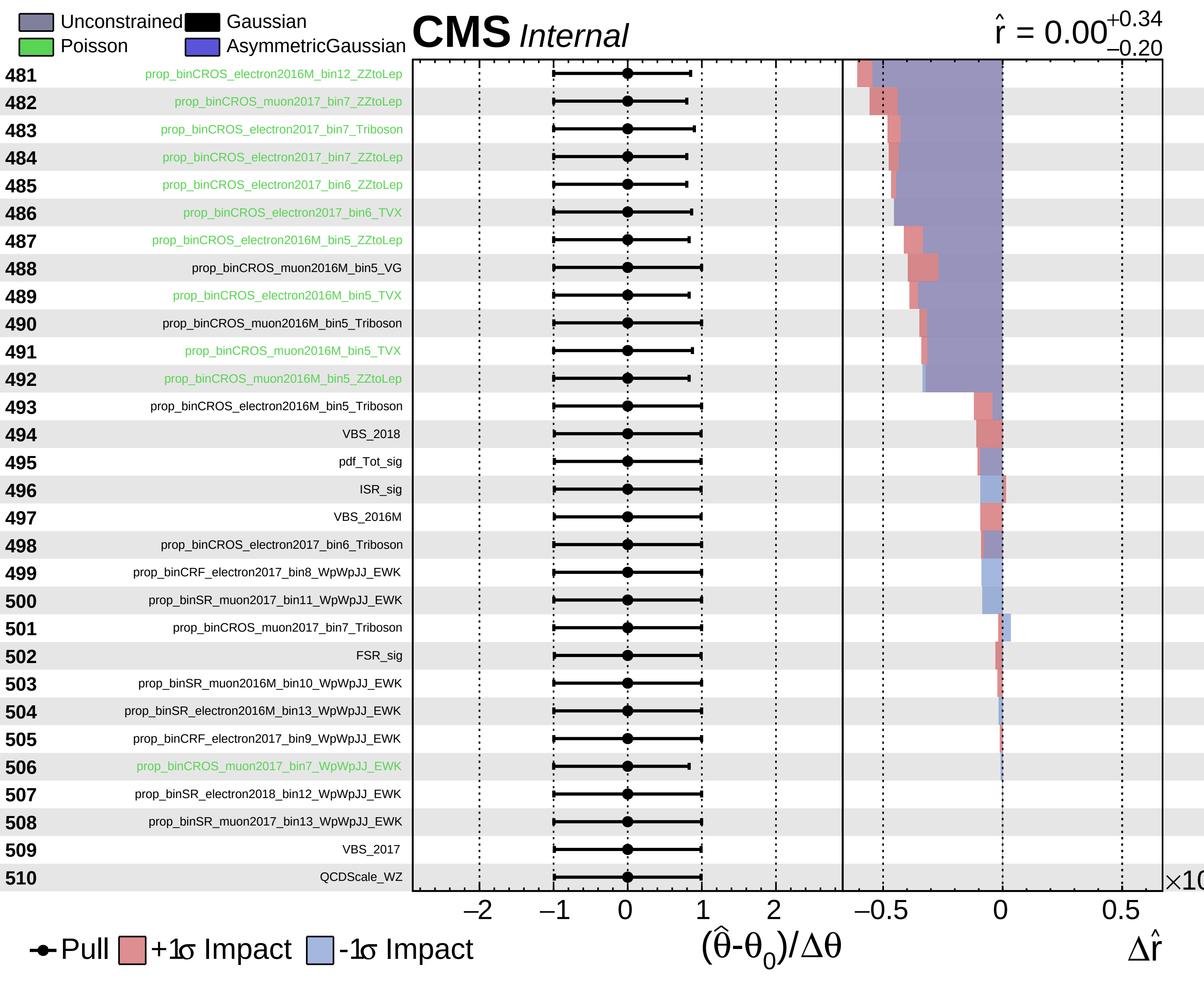


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
391 prop_binCRTT_electron2018_bin9
392 prop_binCRTT_muon2017_bin5
393 prop_binCROS_muon2016M_bin11
394 prop_binCRF_electron2017_bin9_WrongSign
395 prop_binSR_electron2016M_bin0
396 prop_binCRTT_electron2018_bin8
397 lep_2017
398 prop_binCRTT_muon2017_bin12
399 prop_binCRTT_muon2018_bin13
400 prop_binCRF_muon2017_bin12
401 prop_binCROS_muon2016M_bin12
402 prop_binSR_electron2017_bin0
403 prop_binCRF_muon2016M_bin11
404 prop_binCROS_electron2017_bin4
405 prop_binCRTT_electron2017_bin8
406 prop_binCRTT_electron2016M_bin7
407 prop_binCRTT_muon2016M_bin13
408 prop_binCRTT_electron2017_bin4
409 prop_binCRTT_electron2017_bin11
410 prop_binCRTT_electron2017_bin13
411 lep_2018
412 prop_binCRTT_muon2018_bin12
413 prop_binCRTT_electron2016M_bin10
414 prop_binCRTT_muon2016M_bin7
415 prop_binCRTT_electron2016M_bin9
416 prop_binCROS_muon2016M_bin10
417 lep_2016M
418 prop_binCRTT_electron2016M_bin13
419 prop_binCRTT_electron2018_bin6
420 pdf_Tot_VG
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

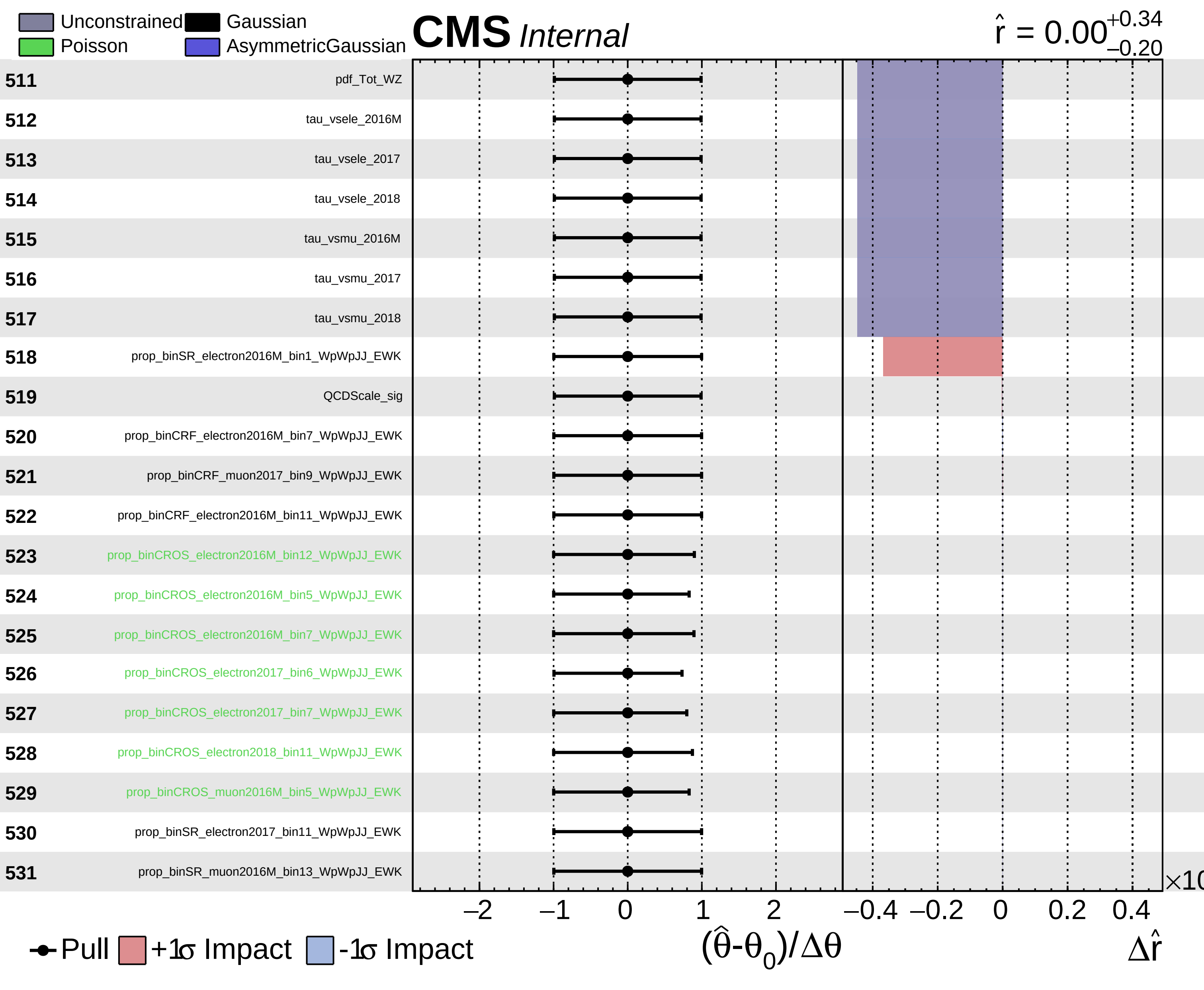


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.20
421 prop_binCRTT_electron2017_bin5
422 prop_binSR_electron2016M_bin13_ZZtoLep
423 prop_binCRTT_electron2018_bin4
424 prop_binCRF_electron2017_bin8_Triboson
425 prop_binSR_electron2016M_bin1_WZ
426 prop_binCRTT_electron2016M_bin2
427 prop_binCROS_electron2016M_bin11
428 prop_binSR_electron2016M_bin1_TTTo2L2Nu
429 prop_binCROS_muon2016M_bin5_TTTo2L2Nu
430 prop_binCRF_electron2017_bin8_ZZtoLep
431 prop_binCRF_muon2016M_bin12
432 prop_binCRF_electron2017_bin9_WpWpJJ_QCD
433 prop_binCRTT_electron2018_bin11
434 pdf_Tot_Triboson
435 prop_binCROS_muon2017_bin7_TTTo2L2Nu
436 prop_binCROS_electron2018_bin11_TTTo2L2Nu
437 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
438 prop_binSR_electron2016M_bin1_VG
439 prop_binCROS_electron2016M_bin12_WrongSign
440 prop_binCRF_electron2016M_bin7_ZZtoLep
441 prop_binCROS_electron2017_bin11
442 prop_binSR_electron2016M_bin13_WrongSign
443 prop_binCROS_muon2017_bin11
444 prop_binCRF_electron2017_bin9_Triboson
445 prop_binCROS_muon2017_bin13
446 prop_binCRF_muon2017_bin9_WrongSign
447 prop_binCRTT_electron2018_bin7
448 prop_binSR_electron2016M_bin1_ZZtoLep
449 prop_binCROS_electron2017_bin7_TTTo2L2Nu
450 prop_binCRF_electron2016M_bin7_TVX
−2
−1
0
1
2
−0.05
0
0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
451 prop_binCROS_electron2016M_bin7_Fake
452 prop_binCROS_electron2016M_bin10
453 prop_binCRF_muon2017_bin9_TVX
454 prop_binCROS_electron2017_bin13
455 prop_binCRF_muon2018_bin13
456 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
457 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
458 prop_binCROS_electron2016M_bin12_Fake
459 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
460 prop_binCRF_muon2017_bin9_Triboson
461 prop_binCROS_electron2017_bin6_TTTo2L2Nu
462 prop_binSR_electron2016M_bin1_TVX
463 prop_binCROS_electron2018_bin11_TVX
464 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
465 prop_binCROS_electron2017_bin7_TVX
466 prop_binCRF_electron2016M_bin7_Triboson
467 prop_binSR_electron2016M_bin1_Triboson
468 prop_binCROS_electron2016M_bin5_VG
469 prop_binCRF_electron2016M_bin7_WpWpJJ_QCD
470 prop_binCROS_electron2018_bin11_Triboson
471 prop_binCRF_electron2017_bin13
472 FES
473 prop_binCROS_electron2018_bin11_Fake
474 prop_binCROS_electron2016M_bin12_Triboson
475 prop_binCROS_electron2018_bin11_ZZtoLep
476 prop_binCROS_electron2016M_bin12_TVX
477 prop_binCROS_electron2016M_bin7_Triboson
478 prop_binCROS_muon2017_bin7_TVX
479 prop_binCROS_electron2016M_bin7_TVX
480 prop_binCROS_electron2016M_bin7_ZZtoLep
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
481 prop_binCROS_electron2016M_bin12_ZZtoLep
482 prop_binCROS_muon2017_bin7_ZZtoLep
483 prop_binCROS_electron2017_bin7_Triboson
484 prop_binCROS_electron2017_bin7_ZZtoLep
485 prop_binCROS_electron2017_bin6_ZZtoLep
486 prop_binCROS_electron2017_bin6_TVX
487 prop_binCROS_electron2016M_bin5_ZZtoLep
488 prop_binCROS_muon2016M_bin5_VG
489 prop_binCROS_electron2016M_bin5_TVX
490 prop_binCROS_muon2016M_bin5_Triboson
491 prop_binCROS_muon2016M_bin5_TVX
492 prop_binCROS_muon2016M_bin5_ZZtoLep
493 prop_binCROS_electron2016M_bin5_Triboson
494 VBS_2018
495 pdf_Tot_sig
496 ISR_sig
497 VBS_2016M
498 prop_binCROS_electron2017_bin6_Triboson
499 prop_binCRF_electron2017_bin8_WpWpJJ_EWK
500 prop_binSR_muon2017_bin11_WpWpJJ_EWK
501 prop_binCROS_muon2017_bin7_Triboson
502 FSR_sig
503 prop_binSR_muon2016M_bin10_WpWpJJ_EWK
504 prop_binSR_electron2016M_bin13_WpWpJJ_EWK
505 prop_binCRF_electron2017_bin9_WpWpJJ_EWK
506 prop_binCROS_muon2017_bin7_WpWpJJ_EWK
507 prop_binSR_electron2018_bin12_WpWpJJ_EWK
508 prop_binSR_muon2017_bin13_WpWpJJ_EWK
509 VBS_2017
510 QCDScale_WZ
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.20}$
511 pdf_Tot_WZ
512 tau_vsele_2016M
513 tau_vsele_2017
514 tau_vsele_2018
515 tau_vsmu_2016M
516 tau_vsmu_2017
517 tau_vsmu_2018
518 prop_binSR_electron2016M_bin1_WpWpJJ_EWK
519 QCDScale_sig
520 prop_binCRF_electron2016M_bin7_WpWpJJ_EWK
521 prop_binCRF_muon2017_bin9_WpWpJJ_EWK
522 prop_binCRF_electron2016M_bin11_WpWpJJ_EWK
523 prop_binCROS_electron2016M_bin12_WpWpJJ_EWK
524 prop_binCROS_electron2016M_bin5_WpWpJJ_EWK
525 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
526 prop_binCROS_electron2017_bin6_WpWpJJ_EWK
527 prop_binCROS_electron2017_bin7_WpWpJJ_EWK
528 prop_binCROS_electron2018_bin11_WpWpJJ_EWK
529 prop_binCROS_muon2016M_bin5_WpWpJJ_EWK
530 prop_binSR_electron2017_bin11_WpWpJJ_EWK
531 prop_binSR_muon2016M_bin13_WpWpJJ_EWK
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$